



COMSATS UNIVERSITY ISLAMABAD, ISLAMABAD CAMPUS

Department of Computer Science

Assignment - 1, Spring 2026

[CLO1]: Characterize the fundamentals of cloud computing along with its service models

[CLO2]: Enumerate virtualization and containerization techniques used in cloud computing

Course: DevOps for Cloud Computing

Class: BCS-7/ BDS-7/ BAI-6

Submission Deadline: March 13, 2026

Total Marks: 10

1- **[CLO1, CLO2]** Amazon Web Services (AWS) is a Public Cloud Platform and is currently the market leader based on various surveys conducted recently. It offers around 200 products and services. In this assignment, you are going to get the flavor of using Cloud Computing Platform and its various service models. Specifically, you are required to build or reuse any previously built web application and deploy it over the Public Cloud, AWS. Since building web applications is out of the scope of this course, it can be any web application with the following requirements:

- It has sufficient functionality ($\approx$ 3 FYP level modules) so that a handsome amount of test cases ($\approx$ 15) can be written in the upcoming assignment
- It uses some Database Server for managing application's data.

You are required to deploy the web application using the following IaaS and PaaS based services offered by AWS:

- Amazon Elastic Compute Cloud (Amazon EC2): An IaaS based service offering the broadest and deepest compute platform
- Amazon Simple Storage Service (Amazon S3): An IaaS based object storage service
- AWS Elastic Beanstalk: A PaaS based service for easy deployments of web applications
- Amazon Relational Database (Amazon RDS): A PaaS based service that allows the creation, operation, management, and scaling of the relational database in the cloud

Upon completion of this assignment, you will be able to:

- Characterize and understand the service models
- Understand and configure servers and applications using virtualized infrastructure
- Configure and deploy application using Infrastructure as a Service (IaaS) model in the cloud
- Setup and deploy application using Platform as a Service (PaaS) model in the cloud

Part-I

In the first part, you will deploy the application using IaaS model and will use EC2 for it. Roughly, the following steps would be needed for the deployment:

- Create an Ubuntu based virtual machine, an EC2 instance
- Add security rules to EC2 instance allowing various accesses (e.g., SSH, HTTP etc.)
- Generate keys pair and use an SSH client to access the EC2 instance.
- Use package manager to install the required servers along with their dependencies and start them
- Configure the permissions
- Set up the Schema and upload your application code, and finally, make your web application up and running

Part-II

In the second part, you will deploy the application using PaaS model and will use S3 to store the application code and Elastic Beanstalk to easily setup the environment for your application. If your application uses a Relational Database, you may also need to set up an RDS instance.

What to submit

This is an individual assignment. Please fill out the following Google form to provide URLs for both the Parts:

<https://forms.gle/jCj9sQxybnURKA3S7>

Here's the link of the response sheet for the above google form to view your responses:

<https://docs.google.com/spreadsheets/d/1yRFY3ap1MvT9iA9TSxml1iwdtobUm1eSb24TGQEbTwM/edit?usp=sharing>

You are also required to submit a *well formatted* report describing your application, documenting all the micro steps along with the screenshots, and listing URLs for both the parts that can be used to access your deployed web application. The report should serve as a manual for someone who wants to deploy the application using EC2 and Elastic Beanstalk.

Evaluation Criteria

Criteria	Marks
Application deployed successfully over EC2 and is up and running	4
Application deployed successfully over Elastic Beanstalk and is up and running	4
Report with screenshots and steps	2